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AI and Computer Vision-based Real-time Quality Control: A Review of Industrial Applications

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Abstract

Computer Vision (CV) provides computers with the ability to perceive, analyze, and understand the content of images and videos. The applications are numerous and range from medical diagnosis to industrial quality control and special effects. CV uses different technologies and techniques to extract relevant information from a large number of images acquired via Machine Vision (MV) components. In industry, CV and MV are combined for trustworthy inspection of high-end products, quality control, and data collection applications. In this article, first, we present the CV and MV with a focus on the convenient artificial intelligence tools that may bused, such as machine & deep learning, and also the common color recognition methods. Second, we consider two industrial applications of artificial vision, especially CV, in real-time quality control to review their scientific valorization based on the primary findings of the studies carried out. Finally, we critically analyzed their findings to identify shortcomings, make recommendations, and open up new horizons for future research.

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Keywords: Computer vision; Artificial intelligence; AI; Machine learning; ML; Deep learning; DL; Machine vision; Real-time quality control; Industry 4.0; Quality 4.0.

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1. Introduction

Nowadays, Computer Vision (CV) techniques continue gaining unprecedented importance in our daily life applications. Nevertheless, according to the latest Gartner Hype Cycle 2023 [1], especially concerning Artificial Intelligence (AI) tools, CV is closely reaching the ultimate productivity plateau. Therefore, it will certainly experience great expansion in the field of industry. Currently, CV already has an important role in the industrial sector, particularly in providing automated inspection capabilities as part of quality control procedures. However, the world of automation is becoming increasingly complex. Industry 4.0, the Internet of Things (IoT), cloud computing, AI, Machine Learning (ML), and many other technologies are presenting vision system users and developers with major challenges in choosing the right and best-suited system for their respective applications [2].

Indeed, with rapid developments in many areas, such as Machine Vision (MV) techniques, CMOS sensors, embedded MV, ML, Deep Learning (DL), robotic interfaces, data transmission standards, and MV capabilities, technology offers benefits to manufacturing on many levels [3]. Indeed, new MV techniques are creating new application potentials. Hyperspectral imaging vision can provide information on the chemical composition of processed materials. Computer imaging allows a series of images to be combined in different ways to bring out details that are impossible to see with conventional MV techniques. Further, polarization cameras can display stresses in materials. Other developments in MV are enabling improved performance, integration, and automation in the manufacturing industry. The degree of integration can range from assisting in manual assembly to full integration into original equipment manufacturers due to the high demands of Industry 4.0.

To ensure that customer satisfaction remains impeccable, the industry must be able to innovate, accelerate, and continuously improve; but above all, it must not lose its assets. If the industry's development is not accompanied by the development of the quality function, this apparent progress risks being to the detriment of product quality, and therefore of the customer experience. We must not ignore “Quality 4.0”, which must be to “Industry 4.0” what quality was to Industry 3.0. Therefore, it is a matter of moving quality controls from paper to the cloud, from the previous technological generation to the current one. The context may change, but the purpose must remain the same. Digital tools now enable organizations to engage more immediately and personally with their contractors, suppliers, and customers—their entire ecosystem. While Keeping in mind the growth that AI can ensure in empowering businesses [4].

For real-time coordination, customer engagement, and streamlined reporting, new technologies have become a guarantee of strong relationships. In addition, by domino effect, overall quality and risk prevention. Where human control is ideal for the qualitative interpretation of a complex and unstructured scene. MV excels in the quantitative measurement of a structured scene due to its speed, accuracy, and repeatability. For instance, on a production line, an MV system can inspect hundreds or even thousands of parts per minute. Further, such a system with appropriate optics and camera resolution can easily inspect objects that are invisible to the naked eye [5].

Afterward, this research study aims to provide the scientific community with an overview of the use of CV & MV in real-world industrial applications. The rest of this article is organized as follows: In Section 2 we will give a background on AI, CV, MV, and the key technologies & techniques used in CV (such as ML, DL, and color detection & measurement) while describing the design of an MV system. Then, we exhibit two industrial applications of CV in Section 3 and discuss their results with critical analysis. The first application is for real-time quality monitoring and the second is for bolts & screws inspection. Finally, we conclude this research work in Section 4 while providing fresh insights into potential future work directions.

2. Background

2.1. Artificial Intelligence (AI)

AI is a subfield of computer science that deals with the development of intelligent agents, or autonomous reasoning, learning, and acting systems. Effective methods for resolving a variety of issues, from game playing to medical diagnosis, have been developed using AI. Indeed, the creation of both ML & DL algorithms has been one of the most significant developments in AI. AI systems can learn from data using ML/DL techniques without having to be explicitly designed or programmed. Today, a wide number of applications employ AI, such as manufacturing,

transportation, healthcare, retail, and customer service. Notably, in manufacturing applications concerned by the current research study, automation of industrial procedures, enhanced quality assurance, and maintenance forecasting are all made possible by AI. For instance, robots equipped with AI can carry out monotonous or hazardous human duties.

2.2. Computer Vision (CV) vs. Machine Vision (MV)

CV is a branch of AI that allows computers and systems to extract meaningful information from digital images or photos, videos, and various other visual inputs — and then act or recommend on that information [6]. Further, CV uses a large amount of data to perform analyses leading to the distinction and then recognition of images. To do so, CV uses one of the most effective types of ML algorithms which is DL; more especially, Convolutional Neural Networks (CNNs). However, MV is a subset of CV, which refers to the study of methods and techniques whereby artificial vision systems can be constructed and usefully employed in practical applications. The introduction of MV to industrial processes is often motivated by a desire to reduce costs by increasing efficiency (and so productivity), reducing errors (and so improving quality), or gathering data. Equally importantly, it may also substitute for an absence of available skilled labor or release workers from dangerous, demanding, or fatiguing industrial activities.

In the past, the definition of the term has been somewhat unclear, however more recently "MV" has, largely de facto, come to be understood as the practical realization of image understanding, or more specifically CV techniques, to help solve practical industrial problems that involve a significant visual component. The more recent emphasis on the marriage between MV and ML, which has so revolutionized the discipline, has been made possible by transformative developments in the field of AI and has served to move some machine vision capabilities closer to that of human vision a long unmet ambition that has existed since the early days of CV as far back as the 1960s [7]. Moreover, we will describe in the next subsection the main components of Machine Vision Systems (MVSs).

2.3. Design of MVSs

MVSs, which serve manufacturing applications like quality control, use image processing and operate on a set of rules and parameters. The main components of an MVS are lighting, optics, image sensors, vision processing, and communications. The lighting illuminates the part to be inspected so that its features stand out and can be precisely seen by the camera (Cf. Fig. 1). The optics acquire an image and then present it to the sensor in the form of light. The sensor of an MVC camera converts this light into a digital light into a digital image that is sent to the processor for analysis.

Therefore, we can split the system into three main components: i) Illumination, ii) Image capturing, and iii) Image processing. Illumination systems play an important role in the quality of the image because it is the light that makes objects visible an appropriate illumination of the visual task allows for an efficient vision. Front- or back-lighting are two categories for lighting configurations. Backlighting is used to highlight the backdrop of the object, whilst front lighting acts as illumination focused on the object enabling improved identification of exterior surface elements of the product. In addition to incandescent and fluorescent lighting, lasers, X-ray tubes, and infrared lights are also utilized [8]. In addition, the samples are seen and generated pictures using image-capturing tools or sensors. Scanners, ultrasound, X-rays, and near-infrared spectroscopy are a few of the tools or sensors used to create pictures.

However, in machine vision, image sensors are often solid-state Charged Coupled Devices (CCD), or cameras, with a few applications utilising thermionic tube devices. With the advent of digital cameras, there is no longer a need for a separate component to convert pictures captured by CCD cameras, photographic cameras, or other sensors to a format that computer processors can understand. Due to their varied resolution, digital camera photos retain their characteristics with minimum noise. Additionally, a computer is used to import and transform the acquired or taken photographs into digital images. Digital images, despite appearing as pictures on the screen, are really numbers that may be read by a computer and transformed into small dots or picture components that represent the actual things [3]. For instance, we can use them as input for authentication systems in different domains such as the ones developed in [9] and [10].

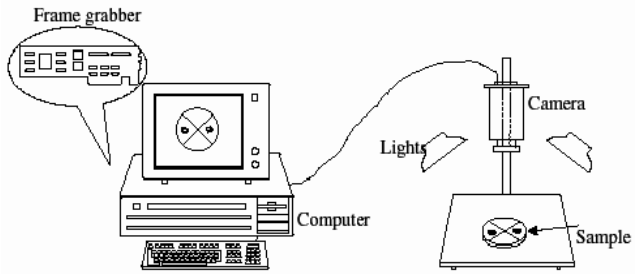

Fig. 1. Machine vision system (MVS) components (Source: [8]).

2.4. Used technologies & techniques in CV

In CV, and through the application of algorithmic models, ML and DL (including CNNs) technologies enable a computer to self-learn & understand the context of the visual data given in the input. So what are these three main technologies (ML, DL, CNNs) that accelerating the development of CV?

2.4.1. Machine Learning (ML)

ML is a branch of AI and computer science that focuses on the development, analysis, and implementation of automated methods that allow a machine to evolve through a learning process. These methods are in practice developed based on algorithms. Indeed, ML algorithms can be broadly categorized into two types: black-box and interpretable. Therefore, when selecting an ML algorithm, it's crucial to take the application's unique requirements into account and strike a balance between “accuracy” and “interpretability”. Indeed, “black-box ML algorithms” are frequently quite accurate, yet they are difficult or impossible to comprehend. Usually, they are developed using sophisticated mathematical models and trained on massive datasets. Black-box algorithms are capable of producing amazing outcomes, but in situations where it is critical to understand how decisions are made, it can be challenging to put your trust in them because of their lack of interpretability. We can list the following black-box algorithms: i) DL algorithms, such as CNNs, recurrent neural networks, and transformers; ii) Support vector machines; iii) random forest; iv) Gradient boosting machines; v) K-nearest neighbor; and vi) Naïve Bayes.

In contrast, “interpretable ML algorithms” are simple to comprehend and interpret by humans. Usually, they are developed using less complex mathematical models using smaller datasets. Despite frequently being less accurate than black-box algorithms, interpretable algorithms are more dependable in applications where interpretability is crucial, such as in financial and medical diagnosis for decision-makers. For instance, this kind of algorithm includes Class Association Rules (CARs) and Regularized Class Association Rules (RCARs) [11][12]. Compared to black-box algorithms, they are frequently less accurate, but they are more suited for applications where it is crucial to understand the reasoning behind decisions such as in finance, security, and medicine. Compared to CARs, RCAR shows high performance and accuracy in several fields, especially in the cybersecurity field when it comes to profiling and preventing security attacks [13][14].

2.4.2. Deep Learning (DL)

DL is a subset of ML in which the tasks are broken down and distributed onto ML algorithms that are organized in consecutive layers. Each layer builds up the output from the previous layer. Together the layers constitute an artificial neural network that mimics the distributed approach to problem-solving carried out by neurons in the human brain [7]. DL includes CNNs as a subset. Therefore, a CNN is a DL algorithm but the opposite is not always the case! Furthermore, CNNs are more often utilized for classification and CV tasks. By dissecting pictures into pixels with labels or tags, a CNN aids an ML or DL model's ability to “look”. Convolutions are a mathematical operation on two functions that results in a third function. It utilizes the labels to do convolutions and predicts what it is “seeing”. Until the predictions start to come true, the neural network conducts convolutions and evaluates the accuracy of its predictions repeatedly. Then, it is identifying or seeing pictures similarly to how people do [6].

2.4.3. Color detection and measurement

To give more insight into how color detection works in image processing, we took the example of CV in food color measurement. Color is the first quality attribute of food evaluated by consumers. We systematically think: "This yellow dessert is going to have a nice vanilla flavor, or this meat is a nice red color that means freshness". As a result, rapid measurement of food color is required in the quality control of food. The human eye distinguishes colors according to the varying sensitivity of different cone cells in the retina to light of different wavelengths. There are three types of color photoreceptor cells (cones) for humans with sensitivity peaks in short (bluish, 420e440 nm), middle (greenish, 530e540 nm), and long (reddish, 560e580 nm) wavelengths [15].

A color sensation no matter how complex can be described using three color components by the eyes There are several ways to code colors, but we present in this article only one to have an idea about color codification Red Green Blue (RGB) is a format of coding of the colors (Cf. Fig. 2). These three colors are the primary colors in additive synthesis. They correspond approximately to the three lengths. of waves to which the three types of cones of the human eye answer. The addition of the three gives white to the human eye. They are used in lighting to obtain all the colors visible to man. They are used today in video, for display on screens & in imaging screens/software [3].

Each of us owns more and more objects, mostly colored cars, phones, jewelry, etc. But have you ever wondered how we judge the colors obtained, how we "measure" them? Indeed, in the industry and for science, the fact of saying that this pair of glasses is a little too blue compared to the other is globally not sufficient. First of all, because each eye is different, one will say that it is too blue, and another will perhaps say that it sees a shade of green on this piece. Well, you should know that in the industry and for science, we prefer to be more rigorous and we can "measure" the color. We can mention three methods of color measurement: i) Visual measurement by trained inspectors; ii) Traditional measurement using colorimeters and spectrometers; and iii) CV measurements [16].

Moreover, for color space and color calibration methods, a specific organization of colors, in combination with physical device profiling, allows for reproducible representations of color, in both analog and digital representations a color space may be arbitrary, with particular colors assigned to a set of physical color swatches and corresponding assigned color names or numbers. A color space is an identifier that tells us how colors will be interpreted and builds those colors around a specific white point [17].

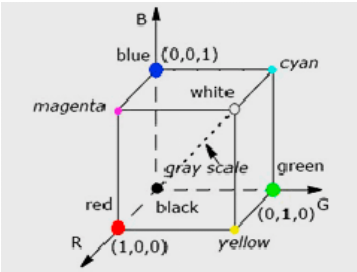

Fig. 2. Red Green Blue (RGB) color coding.

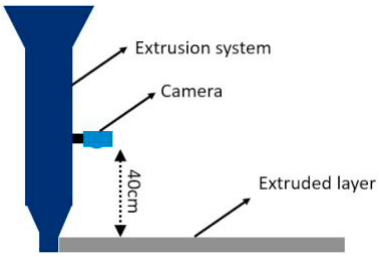

Fig. 3. Schematic layout of the system (Source: [18]).

3. Industrial applications of CV: Result and critical analysis

3.1. Real-time quality monitoring during robotic building construction

In this subsection, we will present the application of CV for real-time quality monitoring during robotic building construction. A laboratory-scale concrete printer is used extrusion to evaluate the performance of the proposed technique through experimentation. Concerning data acquisition and processing, a Logitech 720p camera is used to capture extrusion videos. It is firmly attached to the extruder using a 3D printed mount, such that it is facing the top surface of the printed layer. The distance from the camera lens to the top surface of the extruded layer is fixed at 40cm (Fig. 3). The extrusion videos are processed in real-time by a Raspberry Pi 3 model B [18]. Further, the CV algorithm proposed by Kazemian et al. [18] in Fig. 4 measures the width of the freshly extruded layer and compares it with the target layer width, to detect over-extrusion or under-extrusion conditions. Over-extrusion can be simply identified by the increased layer width, compared to the target width. Under-extrusion, however, could result in three different conditions: reduced layer width, tearing & discontinuity of extruded layer, or zero deposition.

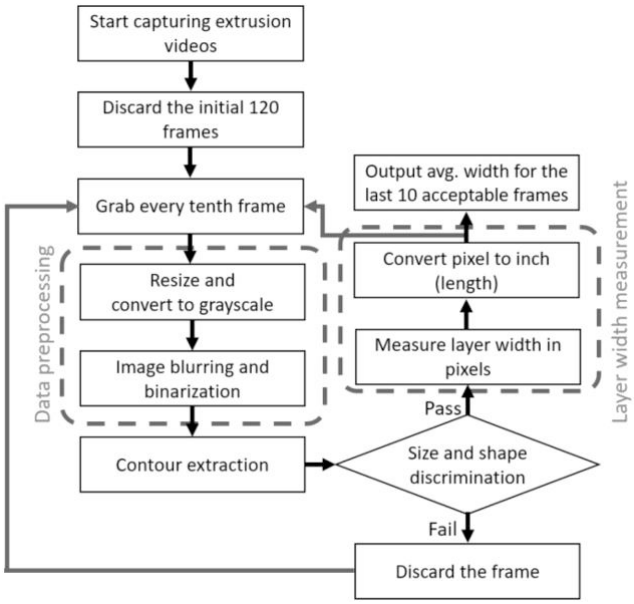

Fig. 4. Proposed computer vision algorithm proposed by A. Kazemian et al. [18].

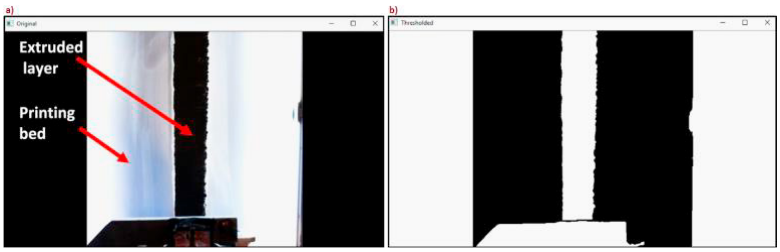

Fig. 5. Prepared picture of the extrusion process: a) Top view; b) Corresponding binary image (Source: [18]).

Fig. 6. Color defect (Source: [20]).

Regarding the video processing algorithm, a shape-based approach is adopted, rather than color and texture-based approaches which usually do not provide reliable results for the detection of slender objects. For each extrusion test, video recording and processing start simultaneously with the extrusion process. The aim is to prepare an easier image for the next step of processing then the picture is blurred using a Gaussian filter, and the image is converted through a binary thresholding method to separate the background (printing bed) and foreground (extruded layer). Then, a binary image is produced in which the "extruded layer" is presented in "white color" and the "background" is produced in "black color" (Cf. **Error! Reference source not found.**). Next step is to detect the contour image through the shape discrimination method, upon finding and verifying a layer contour the width of the layer is measured.

To demonstrate an application of this technique, the CV algorithm proposed by A. Kazemian et al. (Cf. Fig. 4) is utilized to implement a feedback control extrusion system for "Contour Crafting". The objective is to develop an extrusion system capable of automatic adjustment of the extrusion parameters such that a layer with the target width is continuously extruded using any printable mixture, without the need for prior calibration and despite some variations in the printing material. Hence, further research is needed to improve the performance of such a vision-based closed-loop extrusion system, especially concerning the control algorithm. However, as an early-stage study, the obtained results prove the feasibility of developing a vision-based extruder that could automatically print layers of specific dimensions using any printable mixture without the need for prior calibration.

CVSs are always a better solution, as their integrated MVSs allow them to react and adapt to what is happening around and in the output of the process. The obtained results revealed the high precision and responsiveness of the developed extrusion monitoring system under experimental conditions. The use of a single card computer could limit the performance of the system implemented is miniaturized. Connectors, which are often the cause of

problems, are present in small numbers in these systems. This increases reliability but makes repairs difficult or impossible. Perhaps with the use of embedded computers, we could have better results in less time and also make calculations that will improve the quality of the extrusion more by increasing the size of the sample implemented because an embedded system typically controls the physical operations of the machine that it is embedded within, it often has real-time computing constraints. Embedded systems control many devices in common use today. Moreover, for a better quality vision in real-time rendering, we can further consider the three-dimensional (3D) Gaussien splatting for real-time radiance field rendering developed by B. Kerbl et al. [19].

3.2. Inspection of bolts and screws in production line based on CV

In this subsection, we present the result of the study of the application of CV in the inspection of bolts and screws performed by J. Rajan et al. [20]. As a result, AI cameras were used to take a picture of the thing that would be subjected to a vision system inspection. An “analog image” is transformed into a “digital image” by sampling and quantization. Pre-processing is the following phase, which prepares the picture for subsequent processing. The pre-processed picture is used as the starting point for the segmentation process, which streamlines the image representation into a more acceptable format for processing. An Intel Neural Compute Stick (NCS) for edge processing was the acceleration hardware employed in their investigation. The object-oriented programming languages Python and C++ were combined to create the coding language since Python was created to do so. A camera module with "infrared light emitting diodes," which generate infrared light and have an internal vision focus, is used to identify flaws in the item. Although Solid Works was used to create the hardware outcasing.

The camera input data are translated into binary data for subsequent processing and are similar to the visual data that are typically recognized by human eyes. This entire procedure was carried out on a machine equipped with an Intel NCS, a vision sensor, and a microcontroller to store data and carry out the related duty of identifying the flaws. A small and fanless gadget called the Intel Neural computing stick may be used to learn AI programming at the edge. “The microcontroller is enabled for IoT tasks that can be performed based on the defect,” claim John Rajan et al. [20]. Further, the manufacturing line's bolts are taken by the camera as they pass beneath it, analyzed further by the system, and the defective bolts are identified and eliminated. Therefore, we can split defects into three categories: i) Color defect; ii) Orientation defect; and ii) Crack defect.

In fact, issues with the production process or deterioration of the data flow when using the neural model to discover the color defect on the bolt as it moves through the quality testing area are what mostly cause color faults. The contours are created and a signal is delivered by the microcontroller to separate it when the fault is confirmed as a value over the threshold mark (Cf. Fig. 6). However, one of the most frequent Bolt faults is an orientation defect. The misorientations in the bolt can be found using CNN technique [16]. The flaws in the bolt are the disorientations. The problematic bolts are identified as having orientation problems, and they are taken off the assembly line. Some cracks are particularly challenging to find using traditional techniques, which is the third flaw, the crack defect. The data flow as the bolt picture is sent through the neural model to detect the fracture while moving through the quality testing region. To lessen visual distortion, the image was converted to grayscale. To make the bolt visible, blurring is used. Edge detection is used to locate the contours and match them with common data. Finding the crack flaw is a result of data change or mismatching.

To sum up, the active contour method used in this application is a type of segmentation technique that presents some limitations. First, it can often get stuck in local minimum states; this may be overcome by using simulated annealing techniques, which is the cause of computational complexity. Second, they often overlook minute features in the process of minimizing the energy over the entire path of their contours. The third limitation is that accuracy is governed by the convergence criteria, which is used in the energy minimization technique. Thus, higher accuracy requires tighter convergence criteria and therefore longer computation times. The fourth weakness is revealed when the image size is too large, hence this method works slowly. The two last points are that this method is not capable of segmenting the nearest objects and is not so good for video-related operations.

4. Conclusion

In this article, we provided an overview of the use of artificial intelligence technologies (such as machine/deep learning and neural networks) and Computer Vision (CV) in industrial real-time quality control while considering two cases of application (real-time quality monitoring during robotic building construction and inspection of bolts and screws in production line). The results showed how CV is already part of our industrial processes, it improves therefore our perception regarding the quality control of our products in real-time and enables us to more accurately anticipate non-conformities and cut costs. Thereby, a critical analysis of the results of the two CV experiences in industrial applications was conducted to identify the main limitations of the commonly used techniques & methods in such cases. Today, the applications of CV are multiple, whether it is to help an operator identify non-conforming parts, to assist an autonomous mobile robot in its displacements, or to analyze the behavior of objects in a defined space. Indeed, as time goes by, we think it is necessary to make real case analyses near the already industrialized processes to be able to overcome the limitations in future intelligent vision systems. Our future work directions include the investigation of various ML & DL algorithms used in CV and their potential application in the case of manufacturing automotive parts for more effective real-time control quality.

References

[1] Perri, L. (2023, August 17). What's New in Artificial Intelligence from the 2023 Gartner Hype Cycle. *Gartner*. https://www.gartner.com/en/articles/what-s-new-in-artificial-intelligence-from-the-2023-gartner-hype-cycle

[2] Lee, J., Davari, H., Singh, J., & Pandhare, V. (2018). Industrial Artificial Intelligence for industry 4.0-based manufacturing systems. *Manufacturing letters*, **18**, 20-23. https://doi.org/10.1016/j.mfglet.2018.09.002

[3] Wu, D., & Sun, D. W. (2013). Colour measurements by computer vision for food quality control–A review. *Trends in Food Science & Technology*, **29**(**1**), 5-20. https://doi.org/10.1016/j.tifs.2012.08.004

[4] Azmi, M., Mansour, A., & Azmi, C. (Forthcoming in 2023). A Context-aware Empowering Business with AI: Case of Chatbots in Business Intelligence Systems. *Procedia Computer Science*. [Preprint available: https://drive.google.com/file/d/1hWhJrWC-GjUvyib9krCPpFhymP27vwZ_/view?usp=sharing]

[5] Lee, M., Yun, J. J., Pyka, A., Won, D., Kodama, F., Schiuma, G., ... & Zhao, X. (2018). How to respond to the fourth industrial revolution, or the second information technology revolution? Dynamic new combinations between technology, market, and society through open innovation. *Journal of Open Innovation: Technology, Market, and* Complexity, **4**(3), 21. https://doi.org/10.3390/joitmc4030021

[6] IBM, What is computer vision? IBM topics (Accessed on Sept. 25, 2023). https://www.ibm.com/topics/computer-vision

[7] Smith, M. L., Smith, L. N., & Hansen, M. F. (2021). The quiet revolution in machine vision-a state-of-the-art survey paper, including historical review, perspectives, and future directions. *Computers in Industry*, **130**, 103472. https://doi.org/10.1016/j.compind.2021.103472

[8] Narendra, V. G. & Hareesha, K. S. (2010). Prospects of computer vision automated grading and sorting systems in agricultural and food products for quality evaluation. *International Journal of Computer Applications*, **1**(4): 1-9.

[9] Mansour, A., Sadik, M., & Sabir, E. (2015, November). Multi-factor authentication based on multimodal biometrics (MFA-MB) for Cloud Computing. In 2015 IEEE/ACS 12th International Conference of Computer Systems and Applications (AICCSA) (pp. 1-4). IEEE.

[10] Mansour, A., Sadik, M., Sabir, E., & Jebbar, M. (2019). AMBAS: An autonomous multimodal biometric authentication system. International *Journal of Autonomous and Adaptive Communications Systems*, **12**(**3**), 187-217.

[11] Azmi, Mohamed, George C. Runger, and Abdelaziz Berrado (2019). Interpretable regularized class association rules algorithm for classification in a categorical data space. *Information Sciences,* **483**: 313-331.

[12] Azmi, M., & Berrado, A. (2015, October). Towards an interpretable rules ensemble algorithm for classification in a categorical data space. In *2015 10th International Conference on Intelligent Systems: Theories and Applications (SITA)* (pp. 1-6). IEEE.

[13] Eddermoug, N., Mansour, A., Sadik, M., Sabir, E., & Azmi, M. (2023). klm-PPSA v. 1.1: machine learning-augmented profiling and *preventing security attacks in cloud environments. Annals of Telecommunications, 1-27.*

[14] Eddermoug, N., Mansour, A., Sadik, M., Sabir, E., & Azmi, M. (2019, October). Klm-PPSA: Klm-based profiling and preventing security attacks for cloud environments. In *2019 International Conference on Wireless Networks and Mobile Communications (WINCOM)* (pp. 1-7).

[15] Hunt, R. W. G. (1995). The reproduction of Colour, 5th Edition, *Fountain Press*, England. [Available from: https://last.hit.bme.hu/download/firtha/video/Colorimetry/Hunt_Reproduction_of_Color.pdf]

[16] Rezagholi, F., & Hesarinejad, M. A. (2017). Integration of fuzzy logic and computer vision in intelligent quality control of celiac-friendly products. *Procedia Computer Science*, **120**, 325–332. https://doi.org/10.1016/j.procs.2017.11.246

[17] Menesatti, P., Angelini, C., Pallottino, F., Antonucci, F., Aguzzi, J., & Costa, C. (2012). RGB color calibration for quantitative image analysis: The "3D Thin-Plate Spline" warping approach. *Sensors*, **12**(**6**), 7063-7079. https://doi.org/10.3390/s120607063

[18] Kazemian, A., Yuan, X., Davtalab, O., & Khoshnevis, B. (2019). Computer vision for real-time extrusion quality monitoring and control in robotic construction. *Automation in Construction*, **101**, 92–98. https://doi.org/10.1016/j.autcon.2019.01.022

[19] Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics (ToG)*, **42**(**4**), 1-14.

[20] Rajan, A. J., Jayakrishna, K., Vignesh, T., Chandradass, J., & Kannan, T. T. M. (2021). Development of computer vision for inspection of bolt using convolutional neural network. *Materials Today: Proceedings*, **45**, 6931-6935.